COMP 3500SEF · Final Report

Warehouse Logistics Management System
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1. Scope

The goals are traceable order decisions — each order's history is auditable through ledger entries and notifications — and a role-separated workflow. Customers order; staff fulfil; drivers deliver; administrators report. Deployment is local-only: one host, one app (§4); demo seed data only, no real personal data (RK-10).

2. Development process

An increment is a unit of work closed by a review or recorded test run.

Table A. Increments

Increment and date	Trigger	Outcome
Planning, 9–11 Oct 2026	Coursework kickoff	Scope, responsibilities, contracts recorded (register)
17 Oct review	Order and stock functions reviewed	Two defects; fixed as row 3
Confirmation, 26–28 Oct 2026	Row-2 defects: repeat submits, no ledger	Transaction verified (orders tests)
Integrated application, 1 Nov 2026	Role functions integrated	Four workflows runnable end to end
Driver and privacy, 2 Nov 2026	Driver tests outstanding	Both pass — data-minimisation, NFR-03 (privacy tests)
Diagram review, 10–11 Nov 2026	Diagrams diverged	Aligned (review record)
Export simplification, 11 Nov 2026	Admin export took four interactions (issue #15)	Selected-date export in three in-app interface interactions (browser checks)
CI health tests, 11–12 Nov 2026	CI-run record untested	Two regression tests added; suite reached 38 (unit.log)

3. Requirements

FR	Requirement and verification
FR-01	create orders with lines and totals Test
FR-02	enforce lifecycle and stock reservation/release Test (cancel)
FR-03	number orders and shipments; track owners Test
FR-04	show inventory, stock-movement ledger and low-stock alerts Test
FR-05	record and read order notifications Test
FR-06	provide admin aggregates and CSV export Test
FR-07	authenticate and enforce role/object access Test
FR-08	assign shipped orders and record delivery Test

- NFR-01 performance — API p95 (95th percentile) ≤2 s, first contentful paint p95 ≤3 s (measurement). - NFR-02 CI health — every commit ends in green unit, integration and browser runs on every machine (observations). - NFR-03 least privilege — role gates; delivery address only to the assigned driver (privacy tests). - NFR-04 interaction budget — every core operation (§8) takes at most three in-app interface interactions; one interaction is a fill, selection or click (walkthrough). - NFR-05 process quality — every increment closes on a recorded review or test run (§2).

Table B. Risk coverage

RK-01–10, with mitigations, are in risk coverage and risk plan; RK-09–10 are out of scope (§1).

4. System models

Each **order item** references one product; orders hold one or more items and belong to one user; an order has at most one shipment, which records tracking events; products accumulate ledger entries; notifications relate to at most one order.

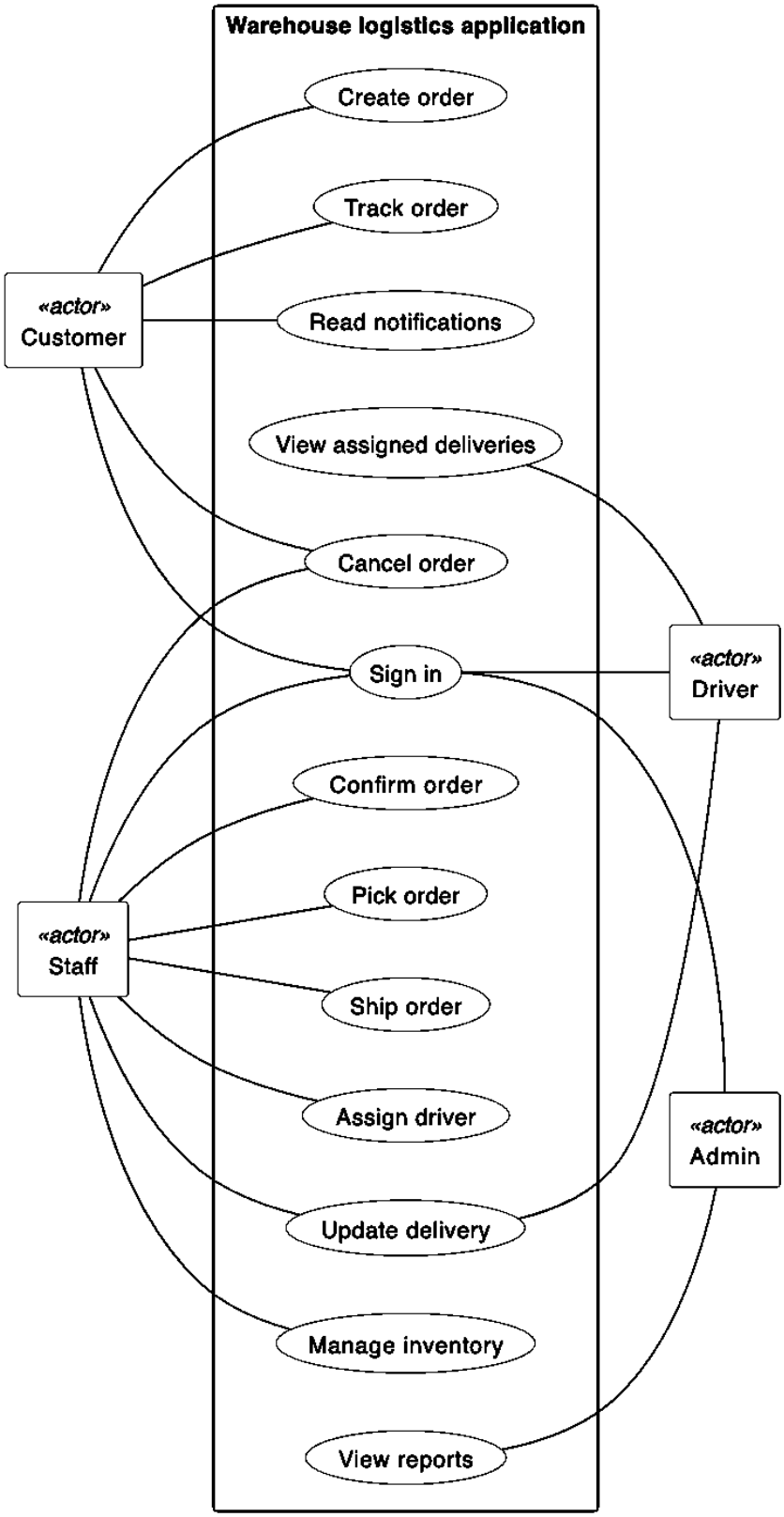


Figure 1. Role functions: four actors, thirteen use cases — what each can do.

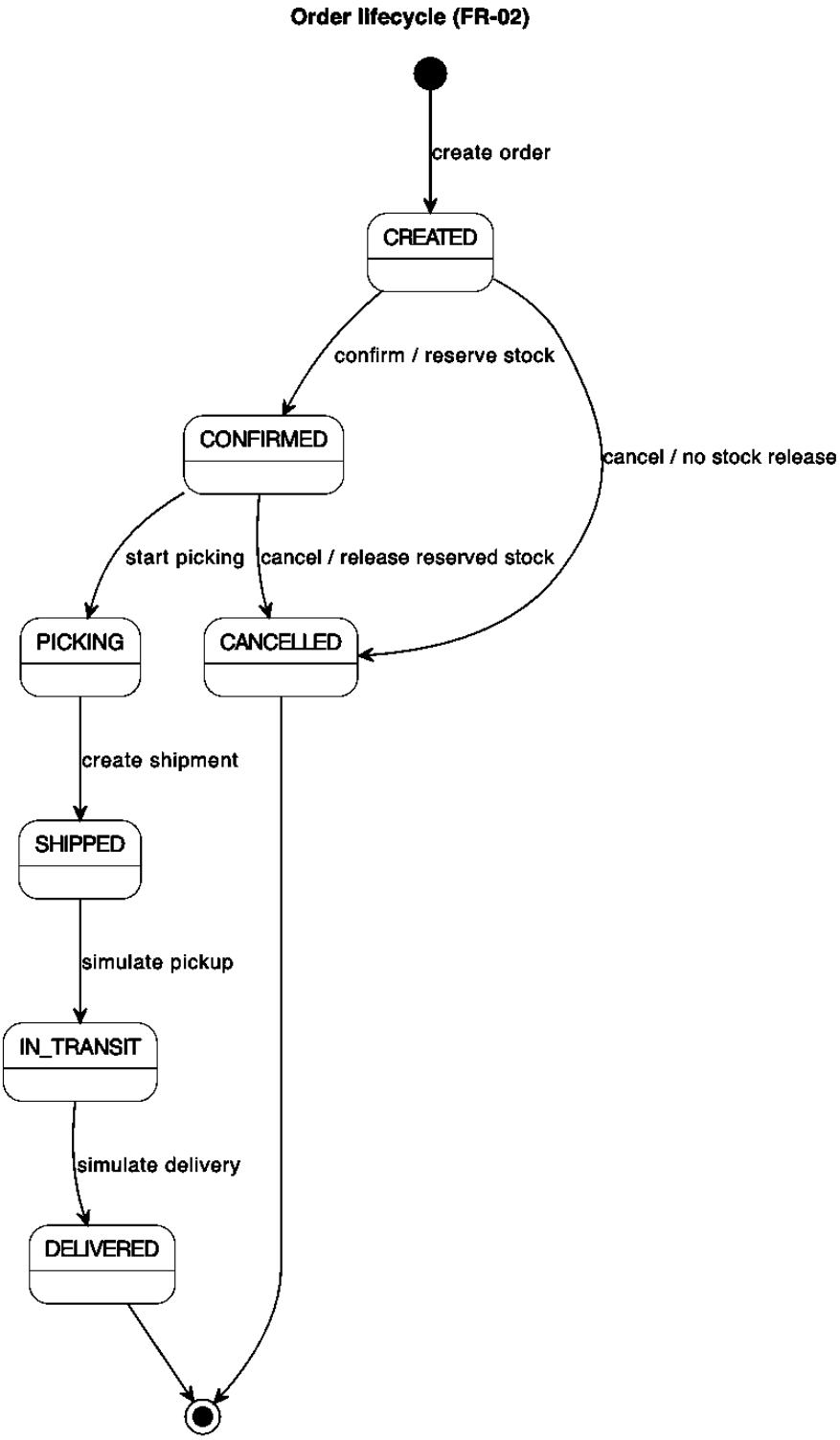


Figure 2. Order transitions and stock effects: each transition reserves, releases or neither.

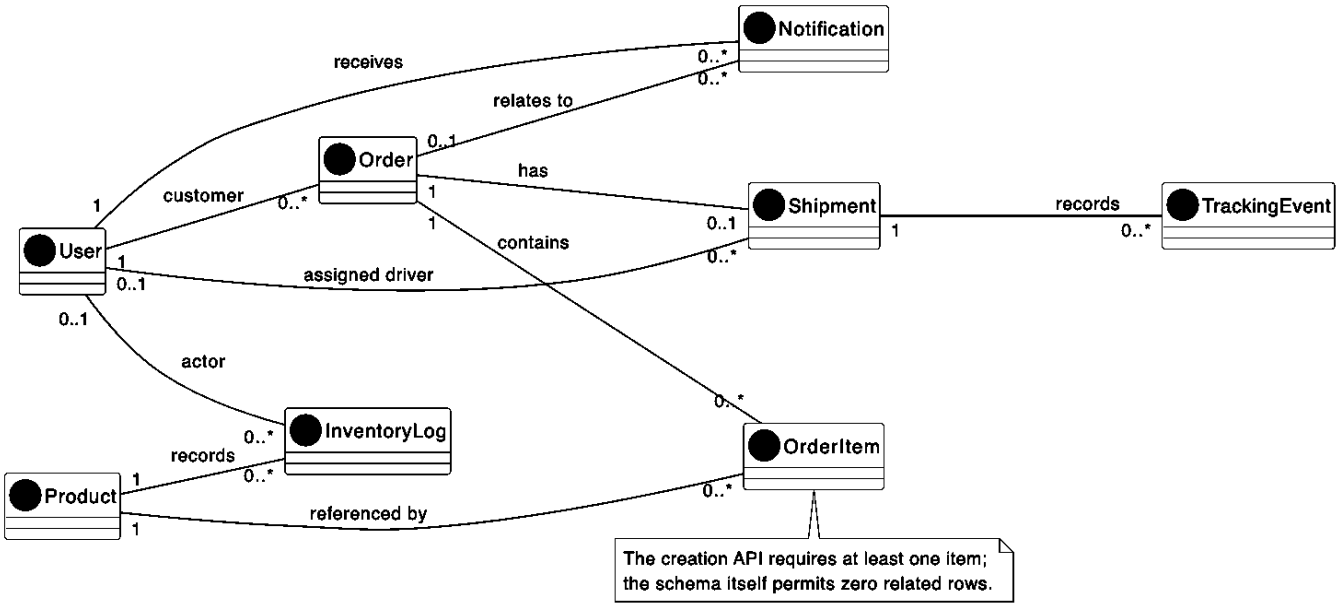


Figure 3. Data relationships: User, Order, OrderItem, Product, Shipment, TrackingEvent, InventoryLog, Notification with multiplicities — the ER model.

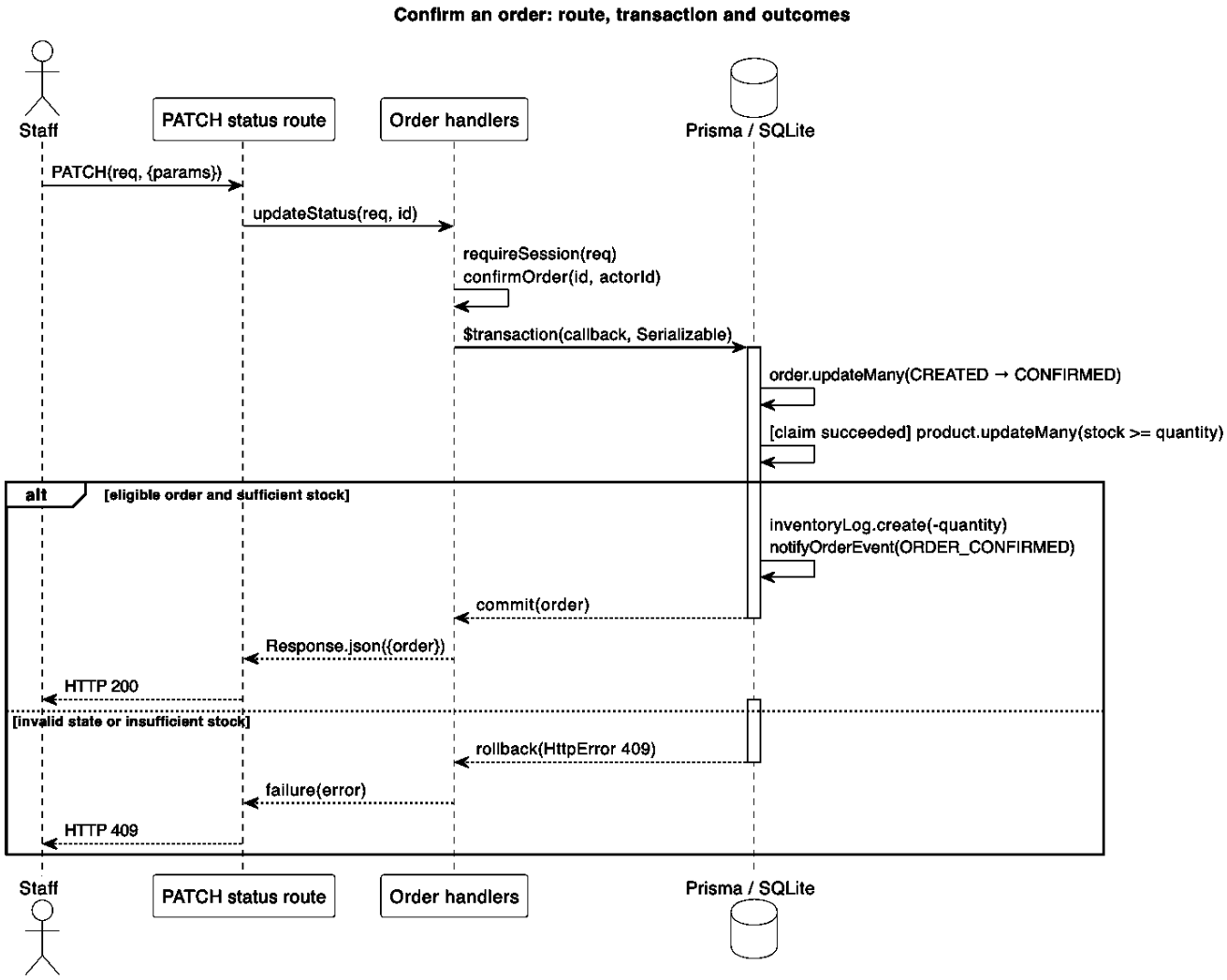


Figure 4. Confirmation transaction: session guard, conditional writes, rollback on failure or short stock.

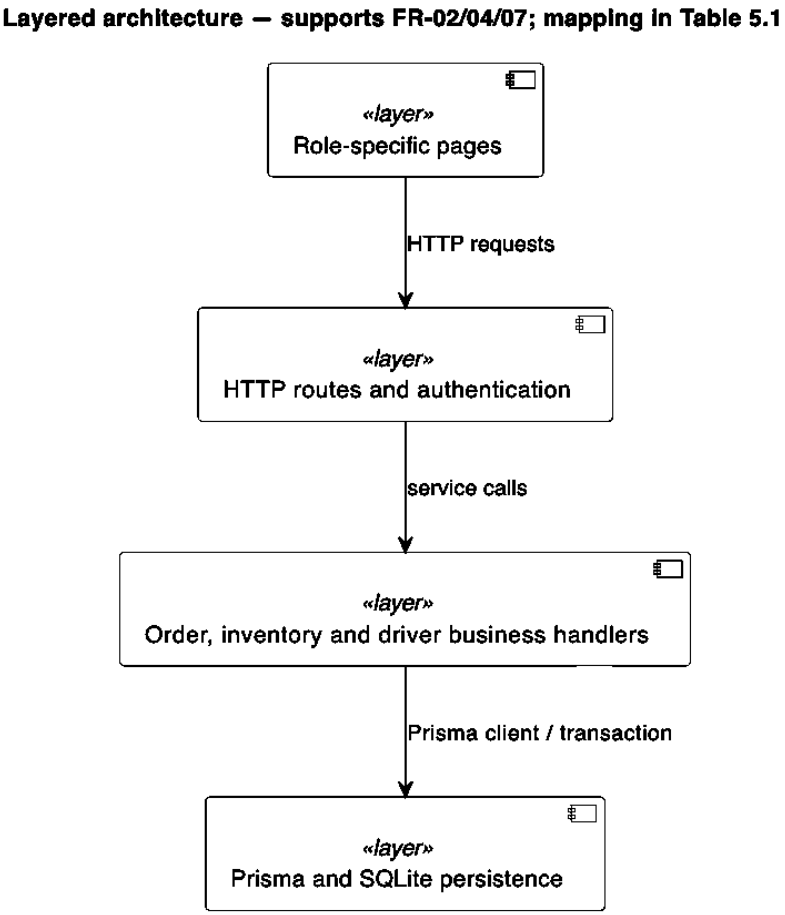


Figure 5. Four layers: pages call routes, routes call handlers, handlers use Prisma/SQLite persistence.

5. Implementation

Confirmation reserves stock. Lifecycle: CREATED → CONFIRMED → PICKING → SHIPPED → IN_TRANSIT → DELIVERED; CANCELLED from CREATED or CONFIRMED.

Cancellation:

Role	State	Effect
Customer	CREATED	cancel, no restock
Customer	CONFIRMED	cancel, restock
Staff	CREATED	rejected
Staff	CONFIRMED	cancel, restock
Customer, staff	PICKING onwards	rejected

A CREATED order stays with the customer; staff have no refuse action in scope (§1); the lifecycle is locked as test constants (tests).

Designs:

Design	Implementation and verification
Preconditions: CREATED, stock sufficient	order-api.ts:250; Orders
Confirmation: atomic; rollback on short stock; repeats rejected (HTTP 409)	order-api.ts:245; Orders
FR-03 numbering and tracking; FR-04 ledger, alerts, staff replenishment	create-order.ts:91, inventory-api.ts:142; Inventory

FR-05/06/07 use the same handler pattern (notify.ts, reports-api.ts, http.ts).

6. Verification

CI records 38 unit, 140 integration checks; 2 integrity checks pin the evidence chain — the demo-data reset is refused unless RESET_DEMO_DATA=YES, and the packaged bundle's SHA-256 manifest must match the pinned source (integrity checks). Logs: integration "tests 140", "pass 140"; unit "Tests 38 passed (38)"; browser "Browser acceptance passed: 26 checks".

Loopback measurement (11 Nov): API p95 3.2–13.6 ms, first contentful paint p95 64–96 ms, against 2 s and 3 s targets; not representative of public networks.

Reviews (Table A rows 2 and 6; checklist inspections) verified requirements and models; row 7's fix was validated through the interface (§8).

7. Evolution

Three reviews drove changes: 17 Oct found the confirmation defect (§5); 10–11 Nov aligned diagrams and traceability and rewrote NFR-01/02 — the daily window (08:00–24:00) became CI-run checks; 11 Nov added the direct export path (issue #15); two regression tests followed as row 8 (36 → 38).

8. Validation

The automated walkthrough exercises eleven core operations — create order; open details; confirm; start picking; ship; assign driver; record pickup; record delivery; filter report; export current; export selected dates — at most three in-app interface interactions each; pickup and delivery are recorded in-app (RK-09). "Browser acceptance passed: 26 checks" — 20 operation checks, 6 cross-cutting (input validation, session handling, interface). No application code changed after this run; row 8 added tests.

9. Feedback and reconciliation

Sources: whole-team reviews; test runs and walkthrough by Oscar and Li.

Table C. Feedback to evidence

Feedback (source, date)	Disposition and evidence
Diagrams diverged (10–11 Nov, corrections)	Corrected
Stock effects under FR-04 (17 Oct, traceability)	Renumbered FR-02
Thin register triggers (10–11 Nov, register, history)	Rewritten
Filter-then-export took four interactions (11 Nov, issue #15)	Combined path not adopted; date selection meets the budget; issue closed
No CI-health test (11 Nov, log)	Suite reached 38
NFR-02 daily availability window (10–11 Nov, revision log)	Replaced by CI-run checks (§3)
NFR-01 hosted measurement (10–11 Nov, measurement)	Scoped to loopback (§6)

10. Evidence

38 unit, 140 integration, 2 integrity (12 Nov), 26 browser checks (11 Nov) — BASELINE. Traceability.